

Lab 4: Hand Gesture Controlled Robot Using Raspberry Pi and ESP32

Course: Robotics

Submission: Group Work

Total Marks: 100

1. Introduction

In this lab, students will build a hand-gesture control system for the robot. A USB camera connected to a Raspberry Pi will capture the user's hand. Python, OpenCV, and MediaPipe will be used to recognize the gesture. The Raspberry Pi will then send a movement command to the ESP32 through serial communication. The ESP32 will control the robot motors according to the received command.

The robot will operate at a fixed speed of 70. The Raspberry Pi is responsible for vision and gesture recognition, while the ESP32 is responsible for receiving commands and controlling the robot movement.

2. Learning Outcomes

By completing this lab, students should be able to:

- Use a Raspberry Pi and USB camera for real-time hand detection.
- Recognize simple hand gestures using MediaPipe hand landmarks.
- Convert detected gestures into robot movement commands.
- Send commands from Raspberry Pi to ESP32 using serial communication.
- Parse received commands on the ESP32 and control robot movement.
- Implement safe stop behavior when no valid gesture is detected.

3. Required Hardware and Software

- Raspberry Pi with Raspberry Pi OS
- USB camera
- ESP32 development board
- Motor driver and robot motors
- USB cable between Raspberry Pi and ESP32 (or UART connection if instructed)
- Python with OpenCV, MediaPipe, and pyserial installed
- Arduino IDE for programming the ESP32

5. Gesture Control Requirements

Students must implement the following five gestures. The movement speed must remain fixed at 70.

Gesture	Robot Action	Command Sent
V sign (index + middle fingers up)	Move Forward	FORWARD,70
Fist	Move Backward	BACKWARD,70
Index finger pointing left	Turn Left	LEFT,70
Index finger pointing right	Turn Right	RIGHT,70
Open palm	Stop	STOP,0

Important: The gestures should be detected clearly before the command is sent. Unknown gestures should not make the robot move.

6. Raspberry Pi Requirements

The Raspberry Pi program must:

- Open the USB camera and display the live video.
- Detect one hand using MediaPipe.
- Draw the detected hand landmarks on the video frame.
- Recognize the five required gestures.
- Display the current detected gesture on the screen.
- Send the corresponding command to the ESP32 through serial communication.
- Send STOP,0 when no hand is detected.
- Avoid repeatedly sending the same command when the gesture has not changed.

7. Serial Communication Requirements

Commands must be sent as text using the following format:

```
FORWARD, 70
BACKWARD, 70
LEFT, 70
RIGHT, 70
STOP, 0
```

Each command should end with a newline character so that the ESP32 can read one complete command at a time.

Example: When the Raspberry Pi detects a V sign, it should send FORWARD, 70 to the ESP32.

8. ESP32 Robot Control Requirements

The ESP32 program must continuously read the serial port and perform the correct movement based on the received command:

- FORWARD,70 -> move the robot forward at speed 70.
- BACKWARD,70 -> move the robot backward at speed 70.
- LEFT,70 -> turn the robot left at speed 70.
- RIGHT,70 -> turn the robot right at speed 70.
- STOP,0 -> stop the robot immediately.

The ESP32 must separate the command and speed value at the comma. **Example:** FORWARD,70 contains the command FORWARD and the speed value 70.

10. Flowchart Requirement

Students must design a flowchart showing the overall system structure, including:

1. System initialization
2. Camera capture
3. Hand detection
4. Gesture recognition
5. Command selection
6. Serial transmission from Raspberry Pi to ESP32
7. ESP32 command parsing
8. Robot movement
9. Stop behavior when no valid hand is detected
10. Continuous looping

11. Demonstration

Each group must demonstrate:

- Live camera view with detected hand landmarks.
- V sign moves the robot forward.
- Fist moves the robot backward.
- Pointing left turns the robot left.
- Pointing right turns the robot right.
- Open palm stops the robot.
- The robot uses a fixed movement speed of 70.
- The robot stops when the hand is removed from the camera.

- Commands sent by the Raspberry Pi can be observed in the terminal.
- The ESP32 correctly receives and executes the commands.

12. Submission Requirements

- Raspberry Pi Python source code (.py file)
- ESP32 source code (.ino file)
- System flowchart diagram
- Written Explanation
- Video demonstration of the complete robot system working